Exp No.: 2
Date : 16/07/20

STUDY OF DIFFERENT TYPES OF NETWORK CABLES

AIM :-

To study and understand different types of network cable, to make your own ethernet cross over cable / straight cable.

DIFFERENT TYPES OF NETWORK CABLE :-

1. Unshielded twisted pair cable
2. Shielded Twisted pair cable
3. Coaxial Cable
4. Fibre Optic Cable.

UTP :-

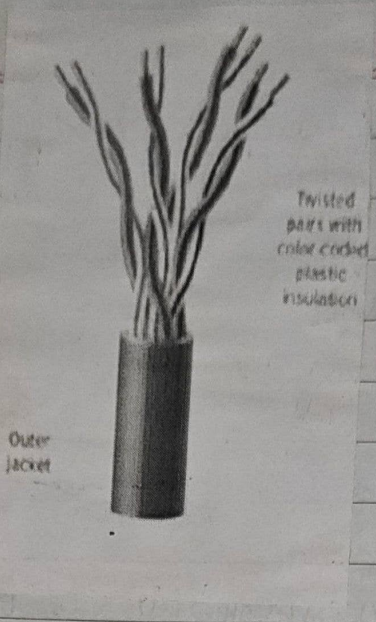
In unshielded twisted pair cable has three categories.

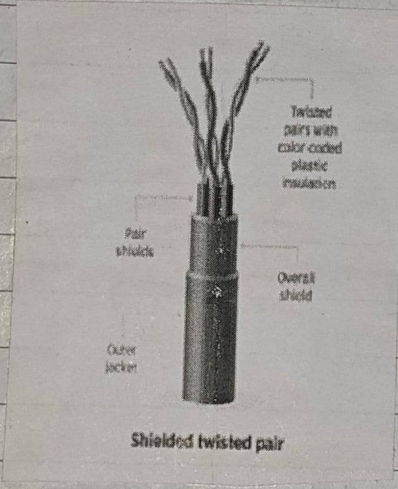
Category 3 has 10 bps data transmission capacity.

Category 5 has upto 100 mbps data transmission capacity.

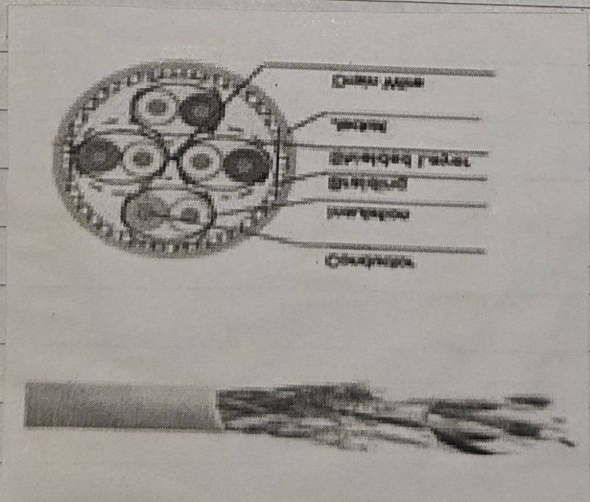
Category 5e has 1Gnps has capacity.

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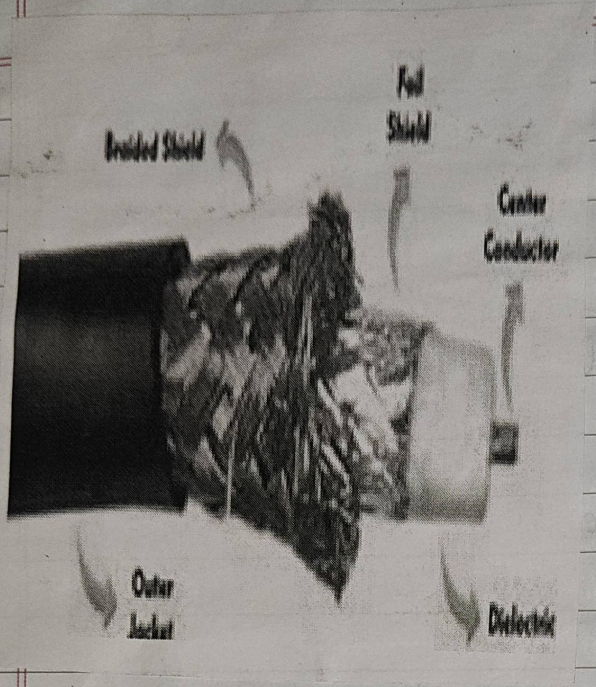


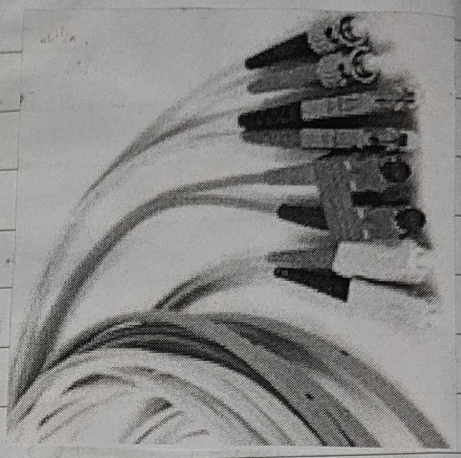
Shielded twisted pair



Cable Type	Category	Maximum Data Transmission	Advantages / disadvantages	Application
UTP	Category 3	10 Mbps	Advantages :- • Cheaper in cost • Easy to install as they have small overall diameter. Disadvantages :- • More prone to EMI	10 Base-T ethernet
	Category 5	upto 100 mbps		• Fast Ethernet • Gigabit ethernet
	Category 5e	1 Gbps		
STP	Category 6, 6a	10 Gbps	Advantages :- • Shielded • Faster than UTP • Less prone to noise	• Gigabit ethernet • Widely used in data centre
SSTP	Category 7	10 Gbps	Disadvantages :- • expensive • Greater installation process	• Gigabit ethernet • 10 G ethernet
Coaxial Cable	RG-6 RG-59 RG-11	10 - 100 Mbps	Advantages :- • High bandwidth • Immune to interface • Versatile Disadvantages :- • cost • Size is bulky	• Speed of signal is soom • High speed internet connection
Fibre Optics cable	Single mode Multi mode	100 Gbps	Advantages :- High Speed High bandwidth High Security Disadvantages :- Expensive	Maximum distance of fibre optics cable is around 100 m

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Cable Type

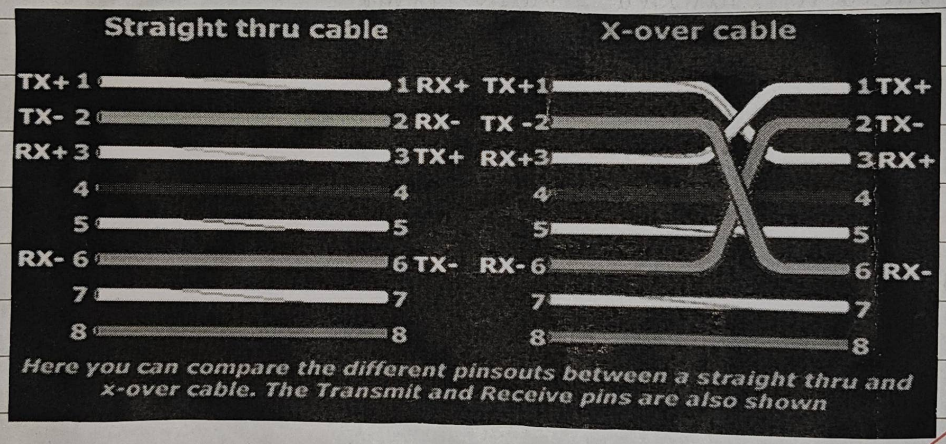
UTP

STP

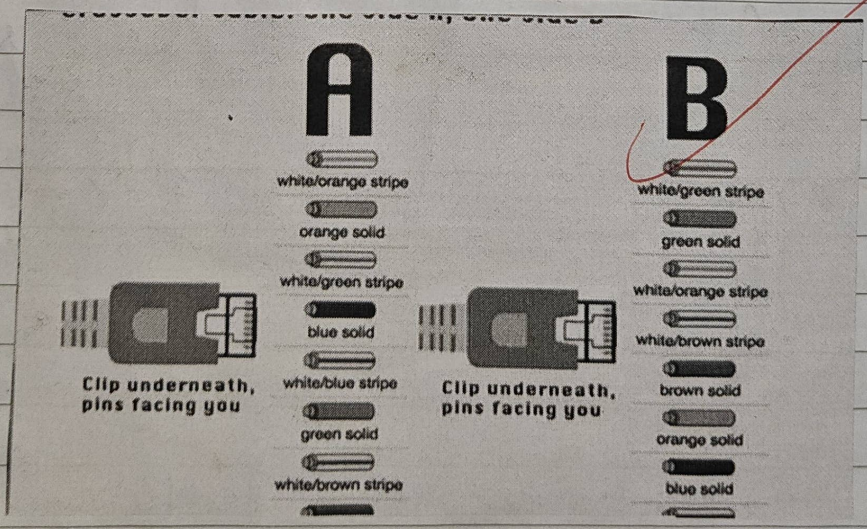
SST

Coax Cab

Fib Opti



Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown



Make your own Ethernet cross-over cable / Straight cable.

Tools and parts needed :-

Ethernet cabling :- Cat 5e is certified for gigabit support but cat 5 cabling works as well.
A crimping tool. This is an all in one networking tool.
Two RJ 45 plugs
Optional two plug shields

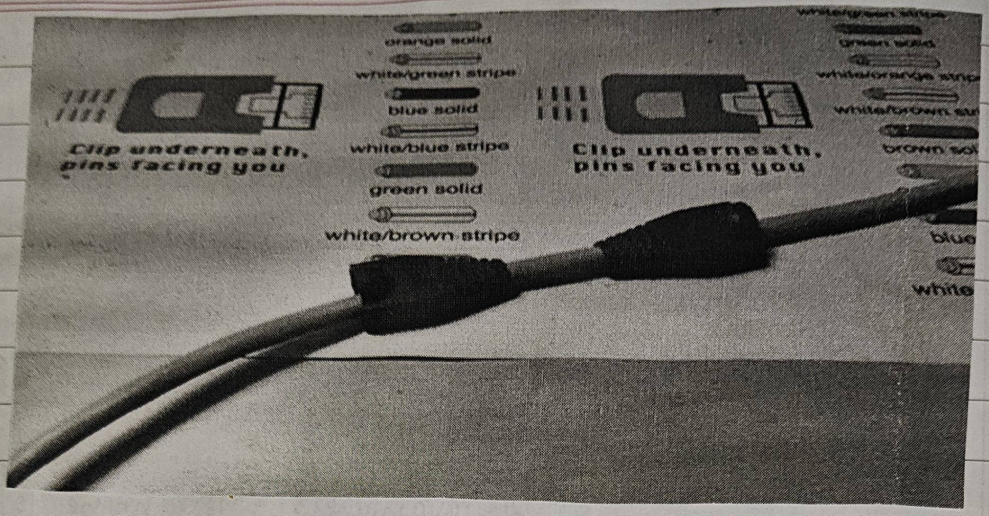
Step 1 :-
To start construction of the device, begin by threading shields on to the cable.

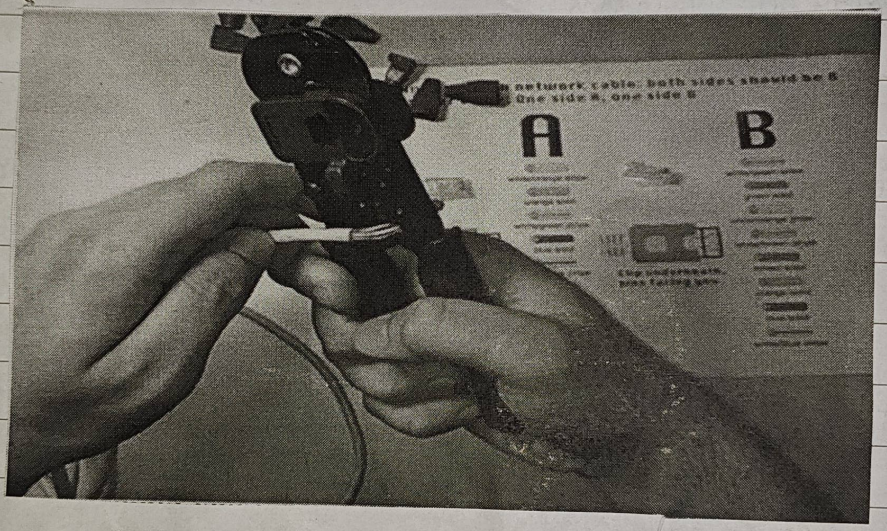
Step 2 :-
Next strip approximately 1.5 cm of the cable shielding from both ends.

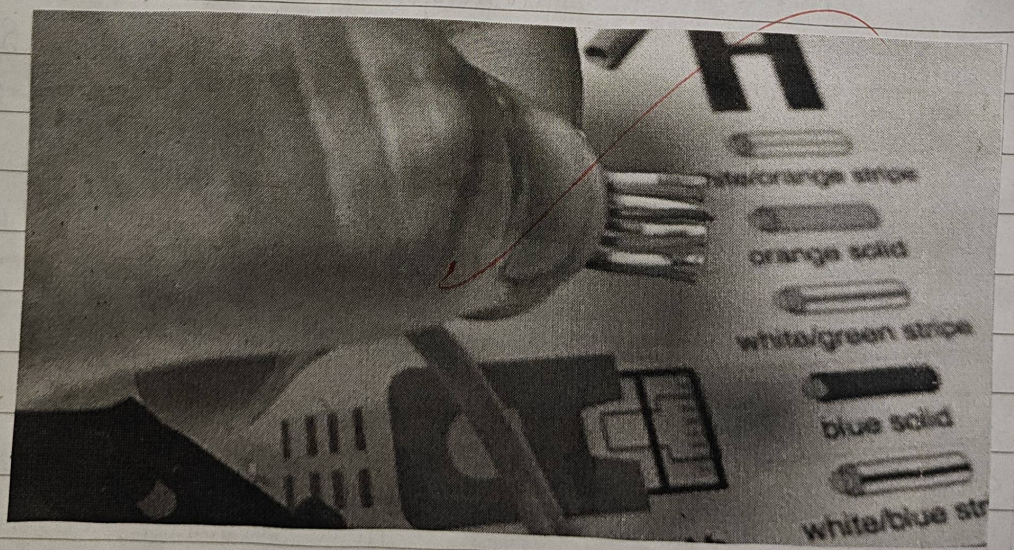
Step 3 :-
After you will need to untangle the wire ; there should be four twisted pairs.

Step 4 :-
Once the order is correct, bunch them to gather in a line and if there are any that stick out farther than others snip them back to create an even level. The difficult aspect is placing the RJ45 plug without messing up the order.

Step 5 :-
Next push the cable right in. The notch at the end to be just over the cable shielding and if it isn't that means that you stripped off too much shielding.


Clip underneath, pins facing you
orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath, pins facing you


A
B


orange solid
white/green stripe
blue solid

Step 6 :-

After the wires are securely sitting inside the plug insert it into the crimp-ing to go and push down.

Step 7 :-

Lastly repeat for the other end using diagram B (to make a crossover cables) / using diagram A (to make a straight through cable).

RESULT :-

Thus, different types of network cables were studied successfully, and straight through and crossover ethernet cables were prepared and tested successfully.

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